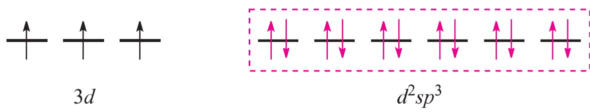
CHEMICAL AND THEORETICAL BACKGROUND

Box 20.1 Valence bond theory: a historical note

In the early use of VB theory, complexes in which the electronic configuration of the metal ion was the same as that of the free gaseous atom were called *ionic complexes*, while those in which the electrons had been paired up as far as possible were called *covalent complexes*. Later, first row metal complexes in which ligand electrons entered 3*d* orbitals (as in $[Fe(CN)_6]^{3-}$) were termed *inner orbital complexes*, and those in which 4*d* orbitals were occupied (as in $[FeF_6]^{3-}$) were *outer orbital complexes*. The various terms, all of which may be encountered, relate to one another as follows:

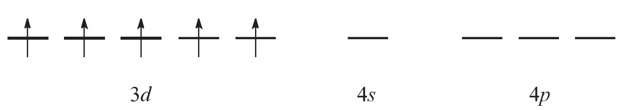
- high-spin complex = ionic complex = outer orbital complex;
- low-spin complex = covalent complex = inner orbital complex.

With the electrons from the ligands included and a hybridization scheme applied for an octahedral complex, the diagram becomes:

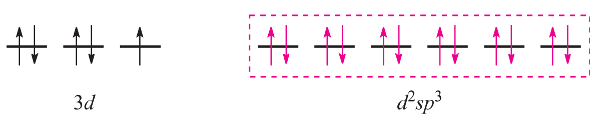


This diagram is appropriate for all octahedral Cr(III) complexes because the three 3*d* electrons always singly occupy different orbitals.

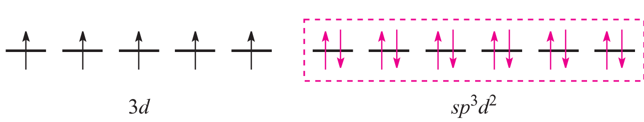
For octahedral Fe(III) complexes, we must account for the existence of both high- and low-spin complexes. The electronic configuration of the free Fe^{3+} ion is:



For a low-spin octahedral complex such as $[Fe(CN)_6]^{3-}$, we can represent the electronic configuration by means of the following diagram where the electrons shown in red are donated by the ligands:

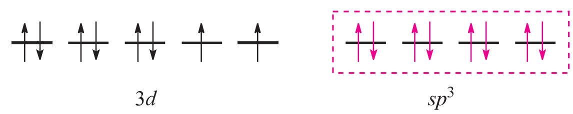


For a high-spin octahedral complex such as $[FeF_6]^{3-}$, the five 3*d* electrons occupy the five 3*d* atomic orbitals (as in the free ion shown above) and the two *d* orbitals required for the sp^3d^2 hybridization scheme must come from the 4*d* set. With the ligand electrons included, valence bond theory describes the bonding as follows leaving three empty 4*d* atomic orbitals (not shown):

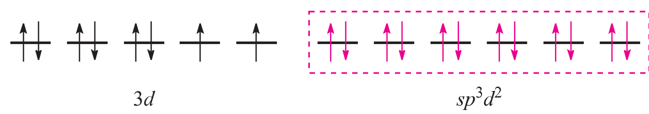


This scheme, however, is unrealistic because the 4*d* orbitals are at a significantly higher energy than the 3*d* atomic orbitals.

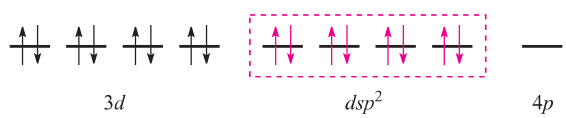
Nickel(II) (d^8) forms paramagnetic tetrahedral and octahedral complexes, and diamagnetic square planar complexes. Bonding in a tetrahedral complex can be represented as follows (ligand electrons are shown in red):



and an octahedral complex can be described by the diagram:



in which the three empty 4*d* atomic orbitals are not shown. For diamagnetic square planar complexes, valence bond theory gives the following picture:



Valence bond theory may rationalize stereochemical and magnetic properties, but only at a simplistic level. It can say *nothing* about electronic spectroscopic properties or about the kinetic inertness (see ***Section 25.2***) that is a characteristic of the low-spin d^6 configuration. Furthermore, the model implies a distinction between high- and low-spin complexes that is actually misleading. Finally, it cannot tell us *why* certain ligands are associated with the formation of high- (or low-)spin complexes. We therefore move on to alternative approaches to the bonding.

20.3 Crystal field theory

A second approach to the bonding in complexes of the *d*-block metals is *crystal field theory*. This is an *electrostatic model* and simply uses the ligand electrons to create an electric field around the metal centre. Ligands are considered as point charges and there are *no* metal–ligand covalent interactions.

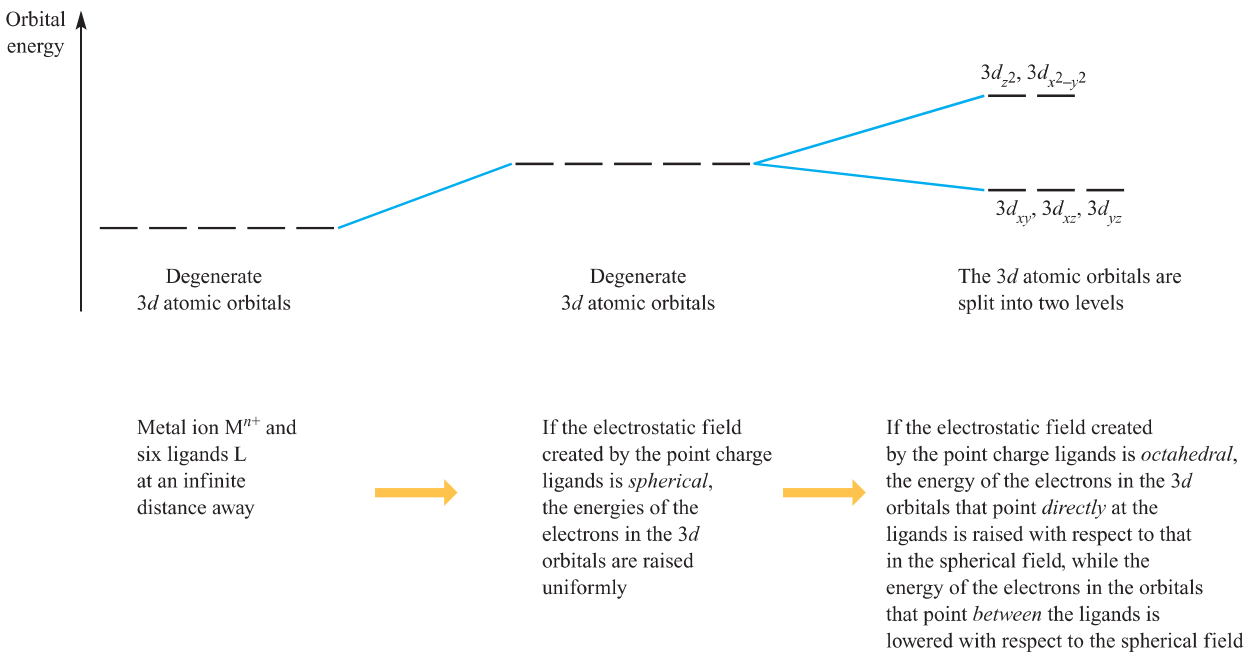


Fig. 20.2 The changes in the energies of the electrons occupying the *d* orbitals of an M^{n+} ion when the latter is in an octahedral crystal field. The energy changes are shown in terms of the orbital energies.

The octahedral crystal field

Consider a first row metal cation surrounded by six ligands placed on the Cartesian axes at the vertices of an octahedron (Figure 20.1a). Each ligand is treated as a negative point charge and there is an electrostatic attraction between the metal ion and ligands. However, there is also a repulsive interaction between electrons in the *d* orbitals and the ligand point charges. *If* the electrostatic field (the *crystal field*) were spherical, then the energies of the five 3*d* orbitals would be raised (destabilized) by the same amount. However, since the d_{z^2} and $d_{x^2-y^2}$ atomic orbitals point *directly at* the ligands while the d_{xy}, d_{yz} and d_{xz} atomic orbitals point *between* them, the d_{z^2} and $d_{x^2-y^2}$ atomic orbitals are destabilized to a greater extent than the d_{xy}, d_{yz} and d_{xz} atomic orbitals (Figure 20.2). Thus, with respect to their energy in a spherical field (the *barycentre*, a kind of 'centre of gravity'), the d_{z^2} and $d_{x^2-y^2}$ atomic orbitals are destabilized while the d_{xy}, d_{yz} and d_{xz} atomic orbitals are stabilized.

> *Crystal field theory* is an electrostatic model which predicts that the *d* orbitals in a metal complex are not degenerate. The pattern of splitting of the *d* orbitals depends on the crystal field, this being determined by the arrangement and type of ligands.

From the O_h character table (***Appendix 3***), it can be deduced (see ***Chapter 4***) that the d_{z^2} and $d_{x^2-y^2}$ orbitals have e_g symmetry, while the d_{xy}, d_{yz} and d_{xz} orbitals possess t_{2g} symmetry (Figure 20.3). The energy separation between them is Δ_{oct} ('delta oct') or $10Dq$. The overall stabi-

CHEMICAL AND THEORETICAL BACKGROUND

Box 20.2 A reminder about symmetry labels

The two sets of *d* orbitals in an octahedral field are labelled e_g and t_{2g} (Figure 20.3). In a tetrahedral field (Figure 20.8), the labels become e and t_2. The symbols t and e refer to the degeneracy of the level:

- a triply degenerate level is labelled *t*;
- a doubly degenerate level is labelled *e*.

The subscript *g* means *gerade* and the subscript *u* means *ungerade*. *Gerade* and *ungerade* designate the behaviour of the wavefunction under the operation of *inversion*, and denote the *parity* (even or odd) of an orbital. The *u* and *g* labels are applicable *only* if the system possesses a centre of symmetry (centre of inversion) and thus are used for the octahedral field, but not for the tetrahedral one.

For more on the origins of symmetry labels: see ***Chapter 4***.

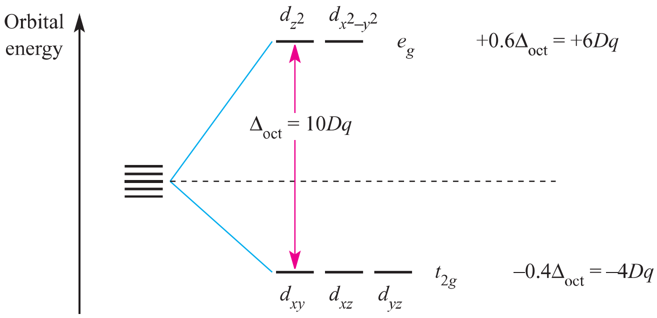
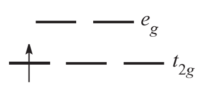


Fig. 20.3 Splitting of the *d* orbitals in an octahedral crystal field, with the energy changes measured with respect to the barycentre.

lization of the t_{2g} orbitals equals the overall destabilization of the e_g set. Thus, orbitals in the e_g set are raised by $0.6\Delta_{oct}$ with respect to the barycentre while those in the t_{2g} set are lowered by $0.4\Delta_{oct}$. Figure 20.3 also shows these energy differences in terms of $10Dq$. Both Δ_{oct} and $10Dq$ notations are in common use, but we use Δ_{oct} in this book.[†] The stabilization and destabilization of the t_{2g} and e_g sets, respectively, are given *in terms of* Δ_{oct}. The magnitude of Δ_{oct} is determined by the *strength of the crystal field*, the two extremes being called *weak field* and *strong field* (equation 20.1).

$$\Delta_{oct}(\text{weak field}) < \Delta_{oct}(\text{strong field}) \qquad (20.1)$$

It is a merit of crystal field theory that, in principle at least, values of Δ_{oct} can be evaluated from electronic spectroscopic data (see *Section 20.6*). Consider the d^1 complex $[Ti(H_2O)_6]^{3+}$, for which the ground state is represented by diagram **20.2** or the notation $t_{2g}{}^1e_g{}^0$.



(20.2)

The absorption spectrum of the ion (Figure 20.4) exhibits one broad band for which $\lambda_{max} = 20\,300\ cm^{-1}$ corresponding to an energy change of $243\ kJ\ mol^{-1}$. (The conversion is $1\ cm^{-1} = 11.96 \times 10^{-3}\ kJ\ mol^{-1}$.) The absorption results from a change in electronic configuration from $t_{2g}{}^1e_g{}^0$ to $t_{2g}{}^0e_g{}^1$, and the value of λ_{max} (see *Figure 20.15*) gives a measure of Δ_{oct}. For systems with more than one *d* electron, the evaluation of Δ_{oct} is more complicated; it is important to remember that Δ_{oct} is an *experimental* quantity.

Factors governing the magnitude of Δ_{oct} (Table 20.2) are the identity and oxidation state of the metal ion and the nature of the ligands. We shall see later that Δ parameters are also defined for other ligand arrangements (e.g. Δ_{tet}). For octahedral complexes, Δ_{oct} increases along the following *spectrochemical series* of ligands; the $[NCS]^-$ ion may

[†] The notation *Dq* has mathematical origins in crystal field theory. We prefer the use of Δ_{oct} because of its experimentally determined origins (see *Section 20.6*).

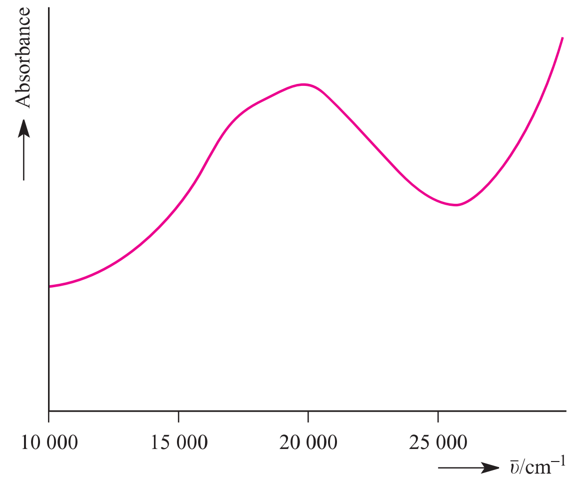


Fig. 20.4 The electronic spectrum of $[Ti(H_2O)_6]^{3+}$ in aqueous solution.

coordinate through the *N*- or *S*-donor (distinguished in red below) and accordingly, it has two positions in the series:

$$I^- < Br^- < [NCS]^- < Cl^- < F^- < [OH]^- < [ox]^{2-}$$
$$\approx H_2O < [NCS]^- < NH_3 < en < bpy < phen < [CN]^- \approx CO$$

weak field ligands ⟶ strong field ligands
increasing Δ_{oct}

The spectrochemical series is reasonably general. Ligands with the same donor atoms are close together in the series. If we consider octahedral complexes of *d*-block metal ions, a number of points arise which can be illustrated by the following examples:

- the complexes of Cr(III) listed in Table 20.2 illustrate the effects of different ligand field strengths for a given M^{n+} ion;
- the complexes of Fe(II) and Fe(III) in Table 20.2 illustrate that for a given ligand and a given metal, Δ_{oct} increases with increasing oxidation state;

Table 20.2 Values of Δ_{oct} for some *d*-block metal complexes.

Complex	Δ / cm^{-1}	Complex	Δ / cm^{-1}
$[TiF_6]^{3-}$	17 000	$[Fe(ox)_3]^{3-}$	14 100
$[Ti(H_2O)_6]^{3+}$	20 300	$[Fe(CN)_6]^{3-}$	35 000
$[V(H_2O)_6]^{3+}$	17 850	$[Fe(CN)_6]^{4-}$	33 800
$[V(H_2O)_6]^{2+}$	12 400	$[CoF_6]^{3-}$	13 100
$[CrF_6]^{3-}$	15 000	$[Co(NH_3)_6]^{3+}$	22 900
$[Cr(H_2O)_6]^{3+}$	17 400	$[Co(NH_3)_6]^{2+}$	10 200
$[Cr(H_2O)_6]^{2+}$	14 100	$[Co(en)_3]^{3+}$	24 000
$[Cr(NH_3)_6]^{3+}$	21 600	$[Co(H_2O)_6]^{3+}$	18 200
$[Cr(CN)_6]^{3-}$	26 600	$[Co(H_2O)_6]^{2+}$	9 300
$[MnF_6]^{2-}$	21 800	$[Ni(H_2O)_6]^{2+}$	8 500
$[Fe(H_2O)_6]^{3+}$	13 700	$[Ni(NH_3)_6]^{2+}$	10 800
$[Fe(H_2O)_6]^{2+}$	9 400	$[Ni(en)_3]^{2+}$	11 500

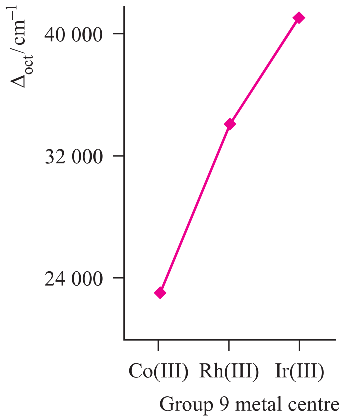


Fig. 20.5 The trend in values of Δ_{oct} for the complexes $[M(NH_3)_6]^{3+}$ where M = Co, Rh, Ir.

- where analogous complexes exist for a series of M^{n+} metals ions (constant n) in a triad, Δ_{oct} increases significantly down the triad (e.g. Figure 20.5);
- for a given ligand and a given oxidation state, Δ_{oct} varies *irregularly* across the first row of the *d*-block, e.g. over the range 8000 to 14 000 cm^{-1} for the $[M(H_2O)_6]^{2+}$ ions.

Trends in values of Δ_{oct} lead to the conclusion that metal ions can be placed in a spectrochemical series which is independent of the ligands:

$$\text{Mn(II)} < \text{Ni(II)} < \text{Co(II)} < \text{Fe(III)} < \text{Cr(III)} < \text{Co(III)} < \text{Ru(III)} < \text{Mo(III)} < \text{Rh(III)} < \text{Pd(II)} < \text{Ir(III)} < \text{Pt(IV)}$$

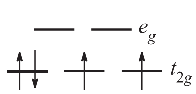
$\xrightarrow{\hspace{6cm}}$
increasing field strength

Spectrochemical series are empirical generalizations and simple crystal field theory *cannot* account for the magnitudes of Δ_{oct} values.

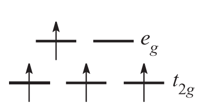
Crystal field stabilization energy: high- and low-spin octahedral complexes

We now consider the effects of different numbers of electrons occupying the *d* orbitals in an octahedral crystal field. For a d^1 system, the ground state corresponds to the configuration $t_{2g}{}^1$ (**20.2**). With respect to the barycentre, there is a stabilization energy of $-0.4\Delta_{oct}$ (Figure 20.3); this is the so-called *crystal field stabilization energy, CFSE.*[†] For a d^2 ion, the ground state configuration is $t_{2g}{}^2$ and the CFSE $= -0.8\Delta_{oct}$ (equation 20.2); a d^3 ion ($t_{2g}{}^3$) has a CFSE $= -1.2\Delta_{oct}$.

$$\text{CFSE} = -(2 \times 0.4)\Delta_{oct} = -0.8\Delta_{oct} \qquad (20.2)$$



(20.3)



(20.4)

For a d^4 ion, two arrangements are available: the four electrons may occupy the t_{2g} set with the configuration $t_{2g}{}^4$ (**20.3**), or may singly occupy four *d* orbitals, $t_{2g}{}^3e_g{}^1$ (**20.4**). Configuration **20.3** corresponds to a low-spin arrangement, and **20.4** to a high-spin case. The preferred configuration is that with the lower energy and depends on whether it is energetically preferable to pair the fourth electron or promote it to the e_g level. Two terms contribute to the electron-pairing energy, P, which is the energy required to transform two electrons with parallel spin in different degenerate orbitals into spin-paired electrons in the same orbital:

- the loss in the *exchange energy* (see ***Box 1.8***) which occurs upon pairing the electrons;
- the coulombic repulsion between the spin-paired electrons.

For a given d^n configuration, the CFSE is the *difference* in energy between the *d* electrons in an octahedral crystal field and the *d* electrons in a spherical crystal field (see Figure 20.2). To exemplify this, consider a d^4 configuration. In a spherical crystal field, the *d* orbitals are degenerate and each of four orbitals is singly occupied. In an octahedral crystal field, equation 20.3 shows how the CFSE is determined for a high-spin d^4 configuration.

$$\text{CFSE} = -(3 \times 0.4)\Delta_{oct} + 0.6\Delta_{oct} = -0.6\Delta_{oct} \qquad (20.3)$$

For a low-spin d^4 configuration, the CFSE consists of two terms: the four electrons in the t_{2g} orbitals give rise to a $-1.6\Delta_{oct}$ term, and a pairing energy, P, must be included to account for the spin-pairing of two electrons. Now consider a d^6 ion. In a spherical crystal field, one *d* orbital contains spin-paired electrons, and each of four orbitals is singly occupied. On going to the high-spin d^6 configuration in the octahedral field ($t_{2g}{}^4e_g{}^2$), no change occurs to the number of spin-paired electrons and the CFSE is given by equation 20.4.

$$\text{CFSE} = -(4 \times 0.4)\Delta_{oct} + (2 \times 0.6)\Delta_{oct} = -0.4\Delta_{oct} \qquad (20.4)$$

For a low-spin d^6 configuration ($t_{2g}{}^6e_g{}^0$) the six electrons in the t_{2g} orbitals give rise to a $-2.4\Delta_{oct}$ term. Added to this is a pairing energy term of $2P$ which accounts for the spin-pairing associated with the two pairs of electrons in excess of the one in the high-spin configuration. Table 20.3 lists values of the CFSE for all d^n configurations in an octahedral crystal field. Inequalities 20.5 and 20.6 show the requirements for high- or low-spin configurations. Inequality 20.5 holds when the crystal field is weak, whereas expression 20.6 is true for a strong crystal field. Figure 20.6 summarizes the preferences for low- and high-spin d^5 octahedral complexes.

$$\text{For high-spin:} \qquad \Delta_{oct} < P \qquad (20.5)$$

$$\text{For low-spin:} \qquad \Delta_{oct} > P \qquad (20.6)$$

We can now relate types of ligand with a preference for high- or low-spin complexes. Strong field ligands such as $[CN]^-$ favour the formation of low-spin complexes, while

[†] The sign convention used here for CFSE follows the thermodynamic convention.

Table 20.3 Octahedral crystal field stabilization energies (CFSE) for d^n configurations; pairing energy, P, terms are included where appropriate (see text). High- and low-spin octahedral complexes are shown only where the distinction is appropriate.

d^n	High-spin = weak field		Low-spin = strong field	
	Electronic configuration	CFSE	Electronic configuration	CFSE
d^1	$t_{2g}{}^1e_g{}^0$	$-0.4\Delta_{oct}$		
d^2	$t_{2g}{}^2e_g{}^0$	$-0.8\Delta_{oct}$		
d^3	$t_{2g}{}^3e_g{}^0$	$-1.2\Delta_{oct}$		
d^4	$t_{2g}{}^3e_g{}^1$	$-0.6\Delta_{oct}$	$t_{2g}{}^4e_g{}^0$	$-1.6\Delta_{oct} + P$
d^5	$t_{2g}{}^3e_g{}^2$	0	$t_{2g}{}^5e_g{}^0$	$-2.0\Delta_{oct} + 2P$
d^6	$t_{2g}{}^4e_g{}^2$	$-0.4\Delta_{oct}$	$t_{2g}{}^6e_g{}^0$	$-2.4\Delta_{oct} + 2P$
d^7	$t_{2g}{}^5e_g{}^2$	$-0.8\Delta_{oct}$	$t_{2g}{}^6e_g{}^1$	$-1.8\Delta_{oct} + P$
d^8	$t_{2g}{}^6e_g{}^2$	$-1.2\Delta_{oct}$		
d^9	$t_{2g}{}^6e_g{}^3$	$-0.6\Delta_{oct}$		
d^{10}	$t_{2g}{}^6e_g{}^4$	0		

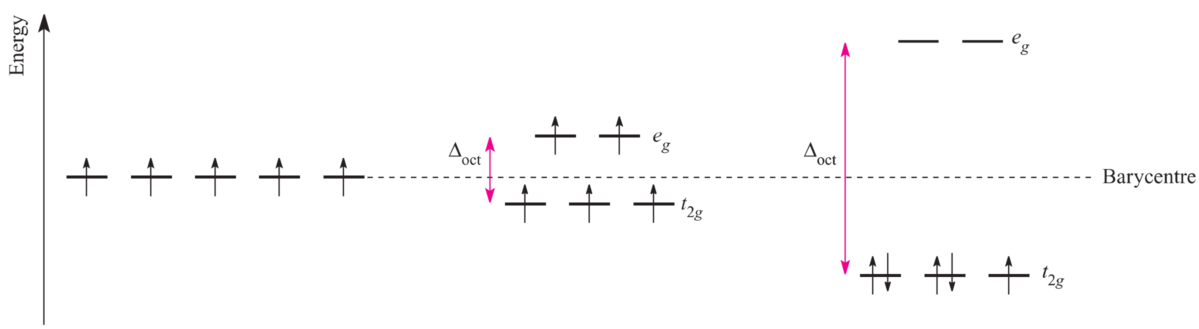


Fig. 20.6 The occupation of the 3d orbitals in weak and strong field Fe^{3+} (d^5) complexes.

weak field ligands such as halides tend to favour high-spin complexes. However, we cannot predict whether high- or low-spin complexes will be formed unless we have accurate values of Δ_{oct} and P. On the other hand, with some experimental knowledge in hand, we can make some comparative predictions: if we know from magnetic data that $[Co(H_2O)_6]^{3+}$ is low-spin, then from the spectrochemical series we can say that $[Co(ox)_3]^{3-}$ and $[Co(CN)_6]^{3-}$ will be low-spin. The only common high-spin cobalt(III) complex is $[CoF_6]^{3-}$.

Jahn–Teller distortions

Octahedral complexes of d^9 and high-spin d^4 ions are often distorted, e.g. CuF_2 (the solid state structure of which contains octahedrally sited Cu^{2+} centres, see *Section 21.12*) and $[Cr(H_2O)_6]^{2+}$, so that two metal–ligand bonds (axial) are different lengths from the remaining four (equatorial). This is shown in structures **20.5** (elongated octahedron) and **20.6** (compressed octahedron).† For a high-spin d^4 ion, one of the e_g orbitals contains one electron while the other is vacant. If the singly occupied orbital is in the d_{z^2}, most of the electron density in this orbital will be concentrated between the cation and the two ligands on the z axis. Thus, there will be greater electrostatic repulsion associated with these ligands than with the other four and the complex suffers elongation (**20.5**). Conversely, occupation of the $d_{x^2-y^2}$ orbital would lead to elongation along the x and y axes as in structure **20.6**. A similar argument can be put forward for the d^9 configuration in which the two orbitals in the e_g set are occupied by one and two electrons respectively. Electron-density measurements

† Other distortions may arise and these are exemplified for Cu(II) complexes in *Section 21.12*.

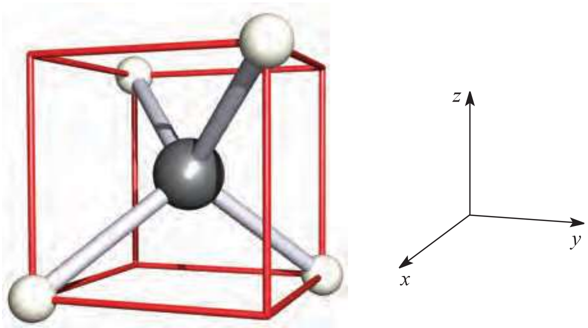
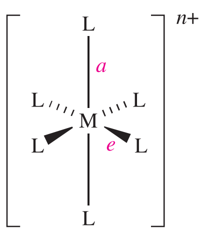


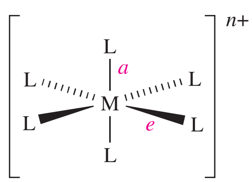
Fig. 20.7 The relationship between a tetrahedral ML_4 complex and a cube; the cube is readily related to a Cartesian axis set. The ligands lie *between* the x, y and z axes; compare this with an octahedral complex, where the ligands lie on the axes.

confirm that the electronic configuration of the Cr^{2+} ion in $[Cr(H_2O)_6]^{2+}$ is *approximately* $d_{xy}{}^1 d_{yz}{}^1 d_{xz}{}^1 d_{z^2}{}^1$. The corresponding effect when the t_{2g} set is unequally occupied is expected to be very much smaller since the orbitals are not pointing directly at the ligands; this expectation is usually, but not invariably, confirmed experimentally. Distortions of this kind are called *Jahn–Teller distortions*.



Bond length $a > e$

(20.5)



Bond length $a < e$

(20.6)

The *Jahn–Teller theorem* states that any non-linear molecular system in a degenerate electronic state will be unstable and will undergo distortion to form a system of lower symmetry and lower energy, thereby removing the degeneracy.

The tetrahedral crystal field

So far we have restricted the discussion to octahedral complexes; we now turn to the tetrahedral crystal field. Figure 20.7 (see also *Figure 4.6*) shows a convenient way of relating a tetrahedron to a Cartesian axis set. With the complex in this orientation, none of the metal d orbitals points exactly at the ligands, but the d_{xy}, d_{yz} and d_{xz} orbitals come nearer to doing so than the d_{z^2} and $d_{x^2-y^2}$ orbitals. For a regular tetrahedron, the splitting of the d orbitals is inverted compared with that for a regular octahedral structure, and the energy difference (Δ_{tet}) is smaller. If all other things are equal (and of course, they never are), the relative splittings Δ_{oct} and Δ_{tet} are related by equation 20.7.

$$\Delta_{tet} = \tfrac{4}{9}\Delta_{oct} \approx \tfrac{1}{2}\Delta_{oct} \qquad (20.7)$$

Figure 20.8 compares crystal field splitting for octahedral and tetrahedral fields; remember, the subscript g in the symmetry labels (see *Box 20.2*) is not needed in the tetrahedral case.

Since Δ_{tet} is significantly smaller than Δ_{oct}, tetrahedral complexes are high-spin. Also, since smaller amounts of energy are needed for an $t_2 \leftarrow e$ transition (tetrahedral) than for an $e_g \leftarrow t_{2g}$ transition (octahedral), corresponding octahedral and tetrahedral complexes often have different colours. (The notation for electronic transitions is given in *Box 20.3*.)

Jahn–Teller effects in tetrahedral complexes are illustrated by distortions in d^9 (e.g. $[CuCl_4]^{2-}$) and high-spin d^4 complexes. A particularly strong structural distortion is observed in $[FeO_4]^{4-}$ (see *structure 21.30*).

The square planar crystal field

A square planar arrangement of ligands can be formally derived from an octahedral array by removal of two *trans*-ligands (Figure 20.9). If we remove the ligands lying along the z axis, then the d_{z^2} orbital is greatly stabilized; the energies of the d_{yz} and d_{xz} orbitals are also lowered, although to a smaller extent. The resultant ordering of the metal d orbitals is shown at the left-hand side of Figure 20.10. The fact that square planar d^8 complexes such as $[Ni(CN)_4]^{2-}$ are diamagnetic is a consequence of the relatively large energy difference between the d_{xy} and $d_{x^2-y^2}$ orbitals. Worked example 20.1 shows an experimental means (other than single-crystal X-ray diffraction) by which square planar and tetrahedral d^8 complexes can be distinguished.

CHEMICAL AND THEORETICAL BACKGROUND

Box 20.3 Notation for electronic transitions

For electronic transitions caused by the absorption and emission of energy, the following notation is used:

Emission: (high energy level) $\rightarrow$ (low energy level)

Absorption: (high energy level) $\leftarrow$ (low energy level)

For example, to denote an electronic transition from the e to t_2 level in a tetrahedral complex, the notation should be $t_2 \leftarrow e$.

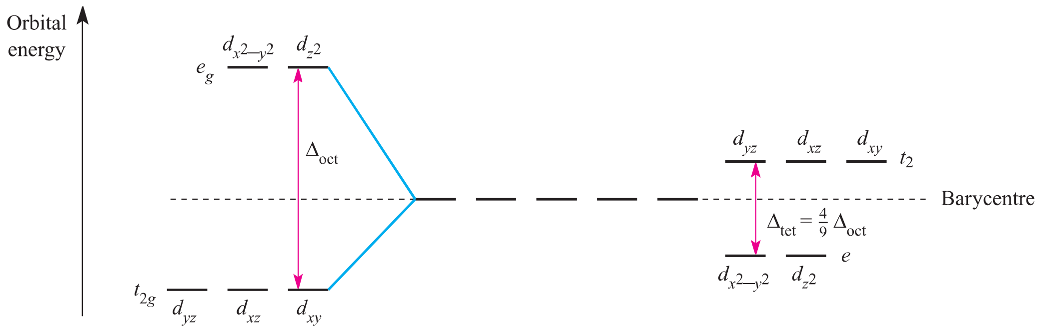


Fig. 20.8 Crystal field splitting diagrams for octahedral (left-hand side) and tetrahedral (right-hand side) fields. The splittings are referred to a common barycentre. See also *Figure 20.2*.

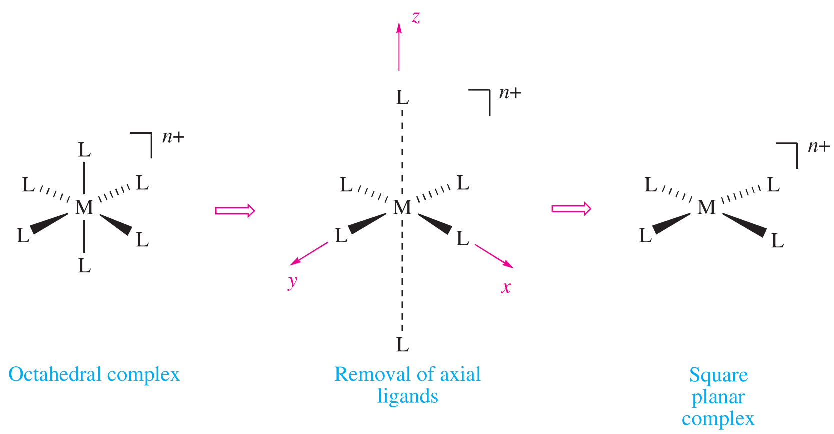


Fig. 20.9 A square planar complex can be derived from an octahedral complex by the removal of two ligands, e.g. those on the z axis; the intermediate stage is a Jahn–Teller distorted (elongated) octahedral complex.

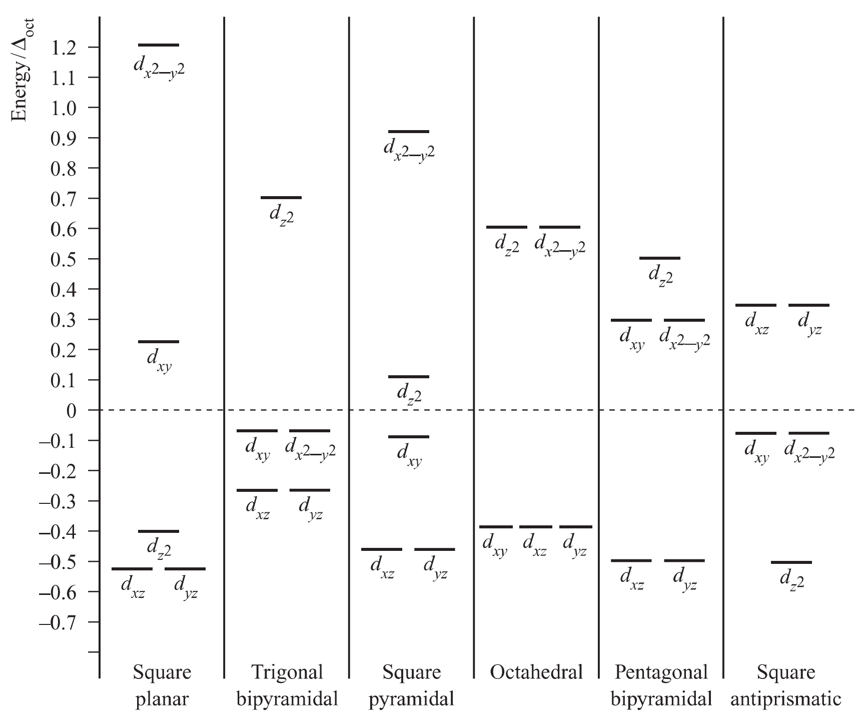
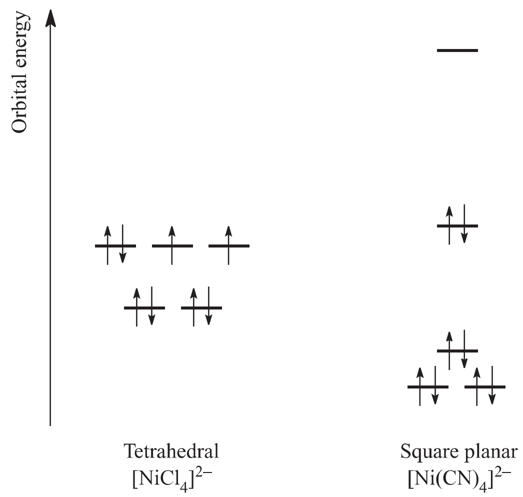


Fig. 20.10 Crystal field splitting diagrams for some common fields referred to a common barycentre; splittings are given with respect to Δ_{oct}. For tetrahedral splitting, see *Figure 20.8*.

Worked example 20.1 Square planar and tetrahedral d^8 complexes

The d^8 complexes $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ are square planar and tetrahedral respectively. Will these complexes be paramagnetic or diamagnetic?

Consider the splitting diagrams shown in Figures 20.8 and 20.10. For $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$, the eight electrons occupy the *d* orbitals as follows:



Thus, $[NiCl_4]^{2-}$ is paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic.

Self-study exercises

No specific answers are given here, but the answer to each question is closely linked to the theory in worked example 20.1.

1. **The complexes $[NiCl_2(PPh_3)_2]$ and $[PdCl_2(PPh_3)_2]$ are paramagnetic and diamagnetic respectively. What does this tell you about their structures?**
2. **The anion $[Ni(SPh)_4]^{2-}$ is tetrahedral. Explain why it is paramagnetic.**
3. **Diamagnetic *trans*-$[NiBr_2(PEtPh_2)_2]$ converts to a form which is paramagnetic. Suggest a reason for this observation.**

Although $[NiCl_4]^{2-}$ is tetrahedral and paramagnetic, $[PdCl_4]^{2-}$ and $[PtCl_4]^{2-}$ are square planar and diamagnetic. This difference is a consequence of the larger crystal field splitting observed for second and third row metal ions compared with their first row congener; Pd(II) and Pt(II) complexes are invariably square planar (but see *Box 20.7*).

Other crystal fields

Figure 20.10 shows crystal field splittings for some common geometries with the relative splittings of the *d* orbitals with respect to Δ_{oct}. By using these splitting diagrams, it is possible to rationalize the magnetic properties of a given complex (see *Section 20.8*). However, a word of caution: Figure 20.10 refers to ML_x complexes containing *like* ligands, and so *only* applies to simple complexes.

Crystal field theory: uses and limitations

Crystal field theory can bring together structures, magnetic properties and electronic properties, and we shall expand upon the last two topics later in the chapter. Trends in CFSEs provide some understanding of thermodynamic and kinetic aspects of *d*-block metal complexes (see *Sections 20.9–20.11* and *25.4*). Crystal field theory is surprisingly useful when one considers its simplicity. However, it has limitations. For example, although we can interpret the contrasting magnetic properties of high- and low-spin octahedral complexes on the basis of the positions of weak- and strong-field ligands in the spectrochemical series, crystal field theory provides no explanation as to *why* particular ligands are placed where they are in the series.

20.4 Molecular orbital theory: octahedral complexes

In this section, we consider a third approach to the bonding in metal complexes: the use of molecular orbital theory. In contrast to crystal field theory, the molecular orbital model considers covalent interactions between the metal centre and ligands.

Complexes with *no* metal–ligand π-bonding

We illustrate the application of MO theory to *d*-block metal complexes first by considering an octahedral complex such as $[Co(NH_3)_6]^{3+}$ in which metal–ligand σ-bonding is dominant. In the construction of an MO energy level diagram for such a complex, many approximations are made and the result is only *qualitatively* accurate. Even so, the results are useful to an understanding of metal–ligand bonding.

By following the procedures that we detailed in *Chapter 4*, an MO diagram can be constructed to describe the bonding in an O_h $[ML_6]^{n+}$ complex. For a first row metal, the valence shell atomic orbitals are 3*d*, 4*s* and 4*p*. Under O_h symmetry (see *Appendix 3*), the *s* orbital has a_{1g} symmetry, the *p* orbitals are degenerate with t_{1u} symmetry, and the *d* orbitals split into two sets with e_g (d_{z^2} and $d_{x^2-y^2}$ orbitals) and t_{2g} (d_{xy}, d_{yz} and d_{xz} orbitals) symmetries, respectively (Figure 20.11). Each ligand, L, provides one orbital and derivation of the ligand group orbitals for the O_h L_6 fragment is analogous to those for the F_6 fragment in SF_6 (see *Figure 4.27*, *equations 4.26-4.31* and accompanying text). These LGOs have a_{1g}, t_{1u} and e_g symmetries (Figure 20.11). Symmetry matching between metal orbitals and LGOs allows the construction of the MO diagram shown in Figure 20.12. Combinations of the metal and ligand orbitals generate six

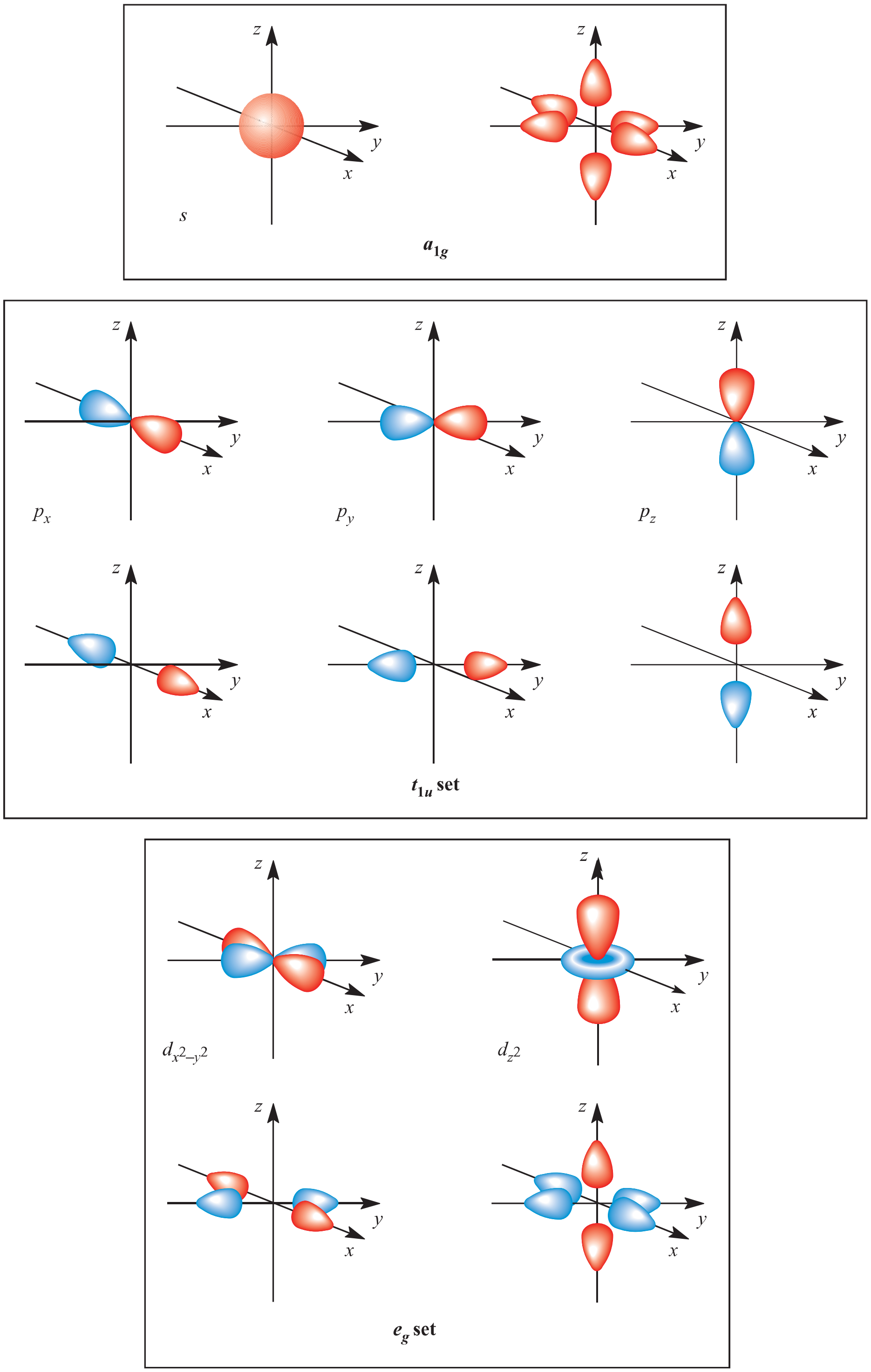


Fig. 20.11 Metal atomic orbitals s, p_x, p_y, p_z, $d_{x^2-y^2}$, d_{z^2} matched by symmetry with ligand group orbitals for an octahedral (O_h) complex with only σ-bonding.

bonding and six antibonding molecular orbitals. The metal d_{xy}, d_{yz} and d_{xz} atomic orbitals have t_{2g} symmetry and are non-bonding (Figure 20.12). The overlap between the ligand and metal s and p orbitals is greater than that involving the metal d orbitals, and so the a_{1g} and t_{1u} MOs are stabilized to a greater extent than the e_g MOs. In an octahedral complex *with no π-bonding*, the energy difference between the t_{2g} and $e_g{}^*$ levels corresponds to Δ_{oct} in crystal field theory (Figure 20.12).

Having constructed the MO diagram in Figure 20.12, we are able to describe the bonding in a range of octahedral σ-bonded complexes. For example: